Quiz 4

[redacted]
[redacted] Apr 9, 2021 [redacted]

Instructions

You have exactly two hours to do this quiz. It is an open-book quiz, and you can use EP2 material given to you this semester (problem-solving solutions, lecture notes, etc). but you are not allowed to use the internet to solve the problems. You can use calculators for math and matrix calculations. If you need help with solving integrals in this quiz - you are allowed to use free integral solving sites, such as **https://www.integral-calculator.com/ (https://www.integral-calculator.com/)** . Needless to say, you also are not allowed to collaborate with anyone or seek help from anyone in solving these problems.

If you have technical problems: immediately contact the KSU help desk and let me know. You can reach me on zoom (**https://ksu.zoom.us/j/95088882288 (https://ksu.zoom.us/j/95088882288)**), discord, and email: **maravin@gmail.com (mailto:maravin@gmail.com)**

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Attempt History

	Attempt	Time	Score
[redacted]			

[redacted]
Submitted Apr 9, 2021 [redacted]
[redacted]

Question 1

3 / 3 pts

The electromotive force that appears in Faraday's law is

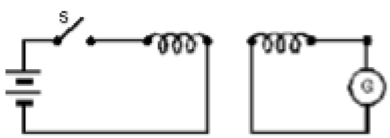
- around a conducting circuit

Correct!

- around the boundary of the surface used to compute the magnetic flux
- throughout the surface used to compute the magnetic flux
- perpendicular to the surface used to compute the magnetic flux
- none of the above

Question 2

3 / 3 pts



Consider a circuit shown above. There is a ferromagnetic bar running through both solenoids, so they share the magnetic fields (if they have them). Find the **true** statement in the list below:

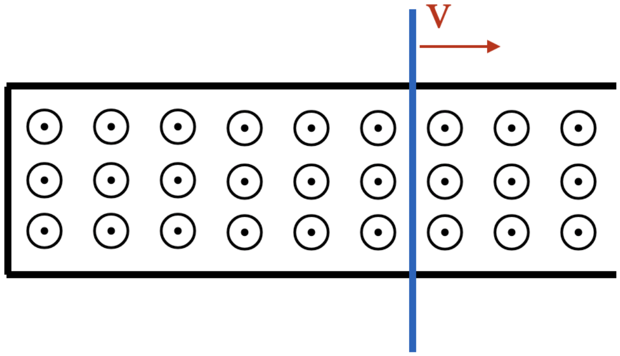
- As long as the switch S is closed, there is a steady reading in the galvanometer G.
- since the two loops are not connected, the current in G is always zero
- the current from the battery goes through the galvanometer G

Correct!

- there is a current in galvanometer G just after the switch S is closed or opened.
- a motional EMF is generated only when the switch S is closed

Question 3

3 / 3 pts

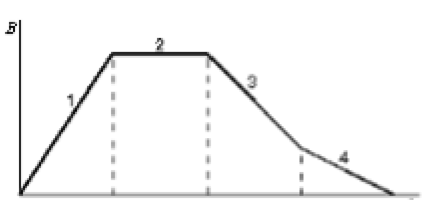


In the figure above you have a rod that lies across two rails in a uniform magnetic field B, as shown. Assume that there is no friction between the rails and the rod. The rod moves to the right with a constant speed *v.* In order for the EMF around the circuit comprised of the rod and rails to be zero, the magnitude of the magnetic field must

- not change
- increase linearly with time
- Correct! decrease linearly with time
- increase quadratically with time
- decrease quadratically with time

Question 4

0 / 3 pts



In the graph above you see the magnitude of a uniform magnetic field B as a function of time in equal periods of time labeled as 1, 2, 3, and 4. For example, during time period 1 the magnetic field increases its magnitude linearly for some period of time, during the time period 2 that lasts the same period of time the magnetic field stays the same etc. The B field is oriented perpendicular to the plane of a conducting loop. Rank the magnitude of the EMF induced in the loop in the four regions, from least to greatest

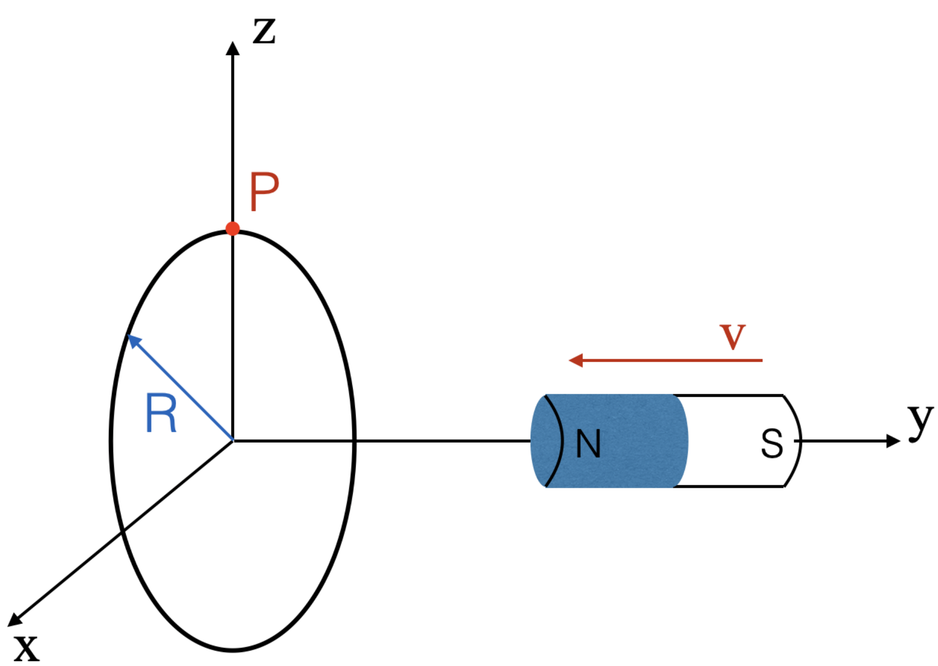
- 1, 2, 3, 4

orrect Answer
- 2, 4, 3, 1
- 4, 3, 1, 2

ou Answered
- 1, 3, 4, 2
- 4, 3, 2, 1

Question 5

3 / 3 pts



In the figure above the magnet is moved toward the loop made from a **non-conductive** material (wood). The loop is located in the x-z plane, centered at the origin, while the magnet is moving along the y-axis. The induced electric field at the top of the ring indicated by point P is directed

- nowhere, as the ring is made of wood

Correct!

- along the positive x-axis direction
- along the negative x-axis direction
- along the positive z-axis direction
- along the negative z-axis direction
- neither of the above

Question 6

3 / 3 pts

Faraday's law states that an induced EMF is proportional to

- the rate of change of the magnetic field
- the rate of change of the electric field

Correct!

- the rate of change of the magnetic flux
- The rate of change of the electric flux
- The rate of change of my EP2 grade during this semester

Question 7

3 / 3 pts

Find the **false** statement in the list below:

- The current across the resistor is independent on the AC power frequency
- Capacitors are short circuit at high frequencies and open circuits at low frequencies

Correct!

- Inductors are short circuits at high frequencies and open circuits at low frequencies
- The current is in phase with the voltage for a resistor
- The current leads the voltage in capacitive circuits

Question 8

3 / 3 pts

While tinkering with your newly purchased oscillograph, you plugged an unknown circuit to an AC voltage supply and found that the current is lagging the voltage. Based on this, you found that the circuit is mostly

- resistive
- Correct! inductive
- capacitive
- burnt and does not work

Question 9

3 / 3 pts

While building your radio receiver, you are making a resonant circuit with an inductance of $2.5\ \mu\mathrm{H}$. To be able to tune to 97.9 MHz radio station, you need the capacitance of

- 3.12 mF
- Correct! 1.06 pF
- 2.12 nF
- $5.14\ \mu\mathrm{F}$
- 2.2 F

Question 10

3 / 3 pts

You measure the oscillation frequency of a given LC circuit to be *f.* You disassemble the circuit and make a new one where the number of inductors and capacitors is doubled. You wired all of them in series to make a new circuit and measure the oscillation frequency to be

- f/4
- f/2
- Correct! f
- 2f
- 4f

Question 11

3 / 3 pts

Consider an LC circuit that has a total energy of 360 μJ and the maximal potential difference on the capacitor to be 15 V. At some point in time, the energy in the capacitor is 40 μJ. What is the potential difference across the capacitor?

- 0 V
- Correct! 5 V
- 10 V
- 15 V

20 V

Question 12

3 / 3 pts

For any RLC series circuit, the impedance will definitely decrease if

- C decreases
- L increases
- L decreases
- R increases
- Correct! R decreases

Question 13

3 / 3 pts

What happens when you half the driving frequency of a series RLC circuit?

- Correct! The capacitive reactance is doubled
- The capacitive reactance is halved
- The impedance is doubled
- The current amplitude is doubled
- The current amplitude is halved

Question 14

3 / 3 pts

If you connect an inductor to the AC power supply, the current lags the voltage by

- 0 (both current and voltage are in phase)
- Correct! one-fourth of a cycle
- one-half of a cycle
- three-fourths of a cycle
- one cycle

Question 15

3 / 3 pts

You have an ideal 1:4 step-down transformer, the primary power is 10 kW and the secondary current is 25 A. The primary voltage is

- 25,600 V
- 3200 V
- Correct! 1600 V
- 400 V
- 50 V

Question 16

3 / 3 pts

You have an ideal transformer that has 100 turns in the primary coil and 800 turns in the secondary. Then, the following statement is **true**

- the power in the primary circuit is less than that in the secondary circuit
- the currents in the two circuits are the same
- the voltages in the two circuits are the same

Correct!

- the primary current is bigger than the one in the secondary circuit
- the frequency in the secondary circuit is eight times that in the primary circuit

Question 17

0 / 3 pts

You connect the primary coil of an ideal 1:8 step-down converter to three 9V batteries connected in series with a 100 Ω resistance. What is the current in the secondary coil?

orrect Answer

- 0 A
- 0.27 A

ou Answered

- 2.16 A
- 0.09 A

0.72 A

Question 18

3 / 3 pts

You connect your power supply for the laptop ranked at 85W to the 120V AC power supply in your house. What is the peak value of the current that flows to your power supply?

- 0.71 A
- Correct! 1.00 A
- 1.41 A
- 0.5 A
- Cannot answer without knowing the impedance of the laptop.

Question 19

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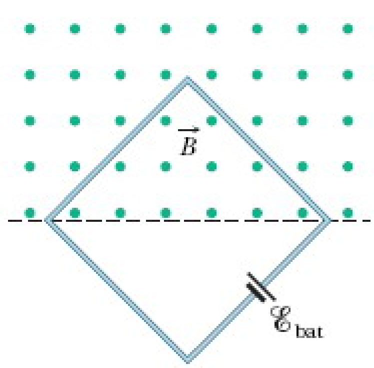

The magnetic flux through the square loop shown in the figure above has the following time dependence Φ_B = 11.0 Wb + 2.0 Wb/s * t, where t is given in seconds. The size of the square's side is 0.1 m. The EMF of a battery is 1.5 V.

A. (3 points) What is the magnitude of the magnetic field at a time t = 2 s?
B. (4 points) What is the net EMF in the circuit?
C. (3 points) What is the direction of the current in the circle? If it changes with time - indicate when.

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Question 20

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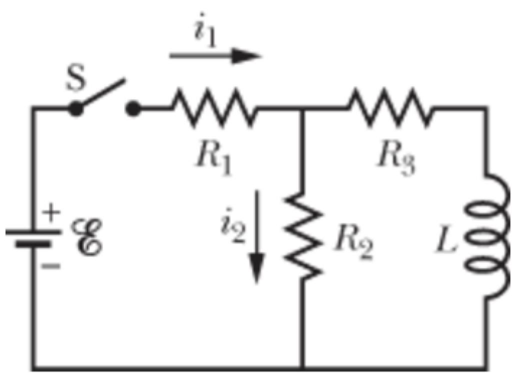

In the figure above, the battery's EMF = 10 V, the resistors R_1 = R_2 = R_3 = 10 kΩ, and L = 2 H. Assume that the currents i_1 and i_2 are positive and are in the direction indicated in the figure (if the currents flow in the opposite direction, indicate them as negative).

A. (4 points) Immediately after the switch S is closed, what are the currents i_1 and i_2?
B. (5 points) After a long time (say a couple of hours), what are the currents i_1 and i_2?
C. (3 points) After the currents are all stabilized, the switch S is re-opened. What are the i_1 and i_2?

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Question 21

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A coil of unspecified shape intercepts its own magnetic flux Φ_B = 40 T m^2 when it draws a current of i = 20 A.

A. (2 points) Show that the inductance of the coil is L = 2 H
B. (3 points) What voltage drop developed across the coil as the current ramped up at a uniform rate from 0 A to 20 A over 10 seconds?
C. (3 points) If the coil were de-energized through a 200 Ω resistor, how long would it take for the current to fall from 20 A to 5 A?
D. (3 points) How much heat energy would be dissipated through the resistor if the coil is completely de-energized?

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Question 22

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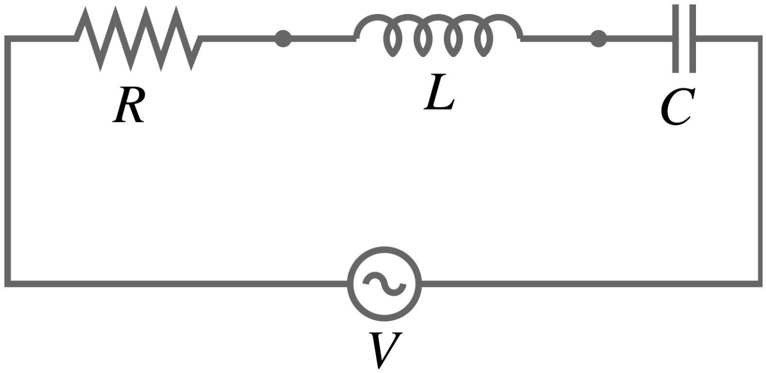

Consider a circuit given above. The circuit is connected to a power outlet with markings 110V and 60Hz. The resistance R = 2 kΩ, L = 2 H, and C = 0.1 μF.

A. (3 points) Find the inductive reactance.
B. (3 points) Find the capacitive reactance.
C. (2 points) Find the impedance of the circuit.
D. (3 points) Find the peak current through the resistor.
E. (2 points) Will the current lag or lead the voltage in this circuit?

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19

a

(+1) (+1)

$$\Phi_B = B\cdot S = B\cdot \frac{1}{2}\cdot a^2,$$ where a is a side of the square

$$B = \frac{2\Phi_B}{a^2} = \frac{2(11\,Wb + 2\,\frac{Wb}{s}\cdot 2\,sec)}{(0.1\,m)^2} = \frac{30\,Wb}{0.01\,m^2} = 3000\,T$$

(+1)

b

From the magnetic field change:

(+1)

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}(11\,Wb + 2\,Wb\cdot t) = -2V$$

From the battery: 1.5V

The Φ_B increases with time (+1)

So the EMF will oppose the change of flux, so the direction of the current will be CW (from RHR) (+1)

(+1)

Opposite to the battery, so $1.5 - 2V = -0.5V$

Note: Do not award points if they just do $1.5V - \frac{d\Phi}{dt}$ without getting the direction correctly.

(c) As the net EMF is in opposite polarity than the battery, the current will flow clockwise.

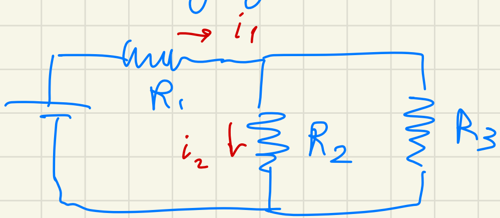
Explanation : (+2)
Correct answer : (+1)

20

(a) Immediately after the switch is closed, no current flows through the inductor (+1), so $i_1 = i_2$ (+1) $= \dfrac{\mathcal{E}}{R_1 + R_2}$ (+1) from Ohm's law

$$i_1 = i_2 = \frac{10\,V}{20\,000\,\Omega} = 0.5\,mA$$ (+1)

(b) Once the inductor is energized it acts as a resistor with (+1) a negligible resistance, so the circuit becomes:



$$i_1 = \frac{\mathcal{E}}{R_1 + \underbrace{\frac{R_2 R_3}{R_1 + R_3}}_{(+1)}} = \frac{10\,V}{10\,k\Omega + 5\,k\Omega}$$ (+1)

$\approx 0.67\,mA$ (+1)

As $R_2 = R_3$, i_1 will be split in 2 equal currents: $i_2 = 0.335\,mA$ (+1)

(c) i_1 will be zero as it is not connected anymore

explanation (+1)

The inductor will continue to pass through the same current as before the switch was opened, so i_2 will reverse direction

(+1)

and $i_2 = -0.335\,mA$.

(+1)

21 (a) From definition: $\Phi = Li \Rightarrow L = \frac{\Phi}{i} = \frac{40W}{20A} = 2H$ (+1) (+1)

(b) Faraday's law: $\mathcal{E} = -\frac{d\Phi}{dt} = -\frac{d}{dt}(Li) = -L\frac{di}{dt} = -2H \cdot \frac{20A}{10\,s} =$ (+1) (+1)

$= -40V$ (+1) OK if value is positive

(c) $i(t) = i_0 e^{-\frac{t}{\tau_{LR}}}$ (+1) $\quad \tau_{LR} = \frac{L}{R} = \frac{2H}{200\Omega} = 0.001\,s$ (+1)

$5A = 20A\, e^{-\frac{t}{0.001s}} \Rightarrow 0.25 = e^{-\frac{t}{0.001s}} \Rightarrow 4 = e^{\frac{t}{0.001s}} \Rightarrow$

$t = 0.001_s \ln 4 = 0.00139\,s$ (+1)

(d) The energy stored in the inductor is

$$PE \overset{(+1)}{=} \frac{Li^2}{2} = \frac{2H \cdot (20A)^2}{2} = 400\,J \quad (+1)$$

When the inductor is deenergized, <u>all</u> that energy is converted into heat, so $Q = 400\,J$ (+1)

22

(a) $f = 60\,Hz \Rightarrow \omega \overset{(+1)}{=} 2\pi f = 376.99\,\frac{rad}{s}$

$X_L \overset{(+1)}{=} \omega L = 377\,\frac{rad}{s} \cdot 2H = 754\,\Omega$ (+1)

(b) $X_C \overset{(+1)}{=} \frac{1}{\omega C} = \frac{1}{377\frac{rad}{s} \cdot 0.1 \cdot 10^{-6} F} = 26500\,\Omega$ (+1)

not f (+1)

(c) $Z \overset{(+1)}{=} \sqrt{R^2 + (X_C - X_L)^2} = \sqrt{(200\Omega)^2 + (26500\Omega - 754\Omega)^2}$

$\simeq 25.8\,k\Omega$ (+1)

(d) 110V is RMS, to get the amplitude we need to convert to peak voltage $V_p = V_{RMS} \cdot \sqrt{2}$ +1

$$i_R = i_C = i_L = \frac{V}{Z} = \frac{110V \cdot \sqrt{2}}{25800\Omega} \simeq 0.006\,A$$

+1 +1

+1 Explanation

(e) The $X_C > X_L$, so the circuit is capacitor dominated, ICE → current will lead the voltage.

+1